A Peculiar Way to Arrange Some Objects Combinatorially

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Abstract

A natural yet relatively advanced question to ask in elementary combinatorics would be the arrangement of some kinds of objects, which may be distinguishable or identical, into containers (called 'groups' hereafter). Then, additional constraints such as the minimum number of each group and some additional composition requirements may be given. This essay attempts to generalise the concepts to a certain extent. After that, a suitable example would be given to illustrate the concepts.

1 An Intuitive Way to State the Problem

Let there be sets where

$$\begin{cases} |A_1| = a_1 \\ |A_2| = a_2 \\ \vdots \\ |A_n| = a_n, \end{cases}$$

where n is an integer in $\mathbb{Z}^+$ (which implies that n can be 1).

Let there be r distinguished groups. The question can be stated vaguely as follows: how many ways are there, preferably with a well-defined general formula or recurrence relation, to arrange the objects into the r labelled groups? Note that the member must be depleted; *id est*, all objects must beong to either one group.

Later on, this article would also aim to discuss some additional constraints, notably the two below:

- Each group must have at least p items; the constraints for p will be built promptly.
- For $n > 1$, each group cannot contain some members from all sets A_1 to A_n. That is, for every group it must be *abstaining* from at least one set A_k (not containing any member originally from it) if and only if $n > 1$. Note that per this definition, the resulting group may or may not contain members exclusively from one set only, which then may or may not be exhaustive.

We will try to formalise everything into definitions, since it would make the path more logical and rigorous. We will start from some elementary counting principles.

We need but the following three principles (or axioms; since proofs would not be available in this article anyway).

Theorem 1 (Addition principle)**.** *If a collection of possibilities is partitioned into disjoint cases $C_1, \ldots, C_m$, then*

$$|C| = \sum_{j=1}^{m} |C_j|.$$

Theorem 2 (Multiplication principle)**.** *If an outcome consists of a sequence of choices, and the number of choices at each stage is fixed independently of the previous choices, then the total number is the product of the numbers of choices.*

Mathematically, this connects to the formula of the probability of the union of two independent events A and B:

$$P(A \cap B) = P(A) \times P(B).$$

Theorem 3 (Decomposition principle)**.** *Because the groups are labelled ('distinguishable'), every distribution is uniquely determined by what goes into the very first group and how the remaining objects are distributed amongst the groups $2, \ldots, r$.*

This rationale is the backbone of our main recurrence.

2 The Cases of the Problem

Assume there is only one set A, so $n = 1$ with a indistinguishable members, and r labelled groups.

Definition 1. For the sake of convenience, define

$$D(a, r)$$

as the number of ways to distribute the aforementioned members.

If the first group receives k members, then the remaining $a-k$ must be put into $r-1$, hence

$$D(a,r)=\sum_{k=0}^{a} D(a-k,r-1).$$

With one single group, one can easily get that

$$D(a,1)=1.$$

To advance, we first need to reiterate the proof of a well-known basic theorem rooted in elementary combinatorics.

Theorem 4 (Stars-and-bars Argument)**.** *To add k nonnegative integers to reach a sum of n, or more generically, to distribute n objects ('stars') into k containers ('bars'), the number of ways is given by $\binom{n+k-1}{k-1}$.*

Generalisation. We have n stars and $k-1$ bars. The minus-one here is given by intuition. Then we are left with a total of $n+k-1$ slots to fill in a row.

Because every unique arrangement of these symbols corresponds to a distinct outcome, to completely determine an arrangement we only need to choose which $k-1$ of the $n+k-1$ total slots contain the *bars*, then the remaining slots automatically be destined to become stars. Then it perfectly matches the definition of combination, so we write

$$\mathrm{C}_{k-1}^{n+k-1}$$

as the answer.

However, because the notation C is very ugly, we write in the form of the binomial coefficient (merely an alternative notation):

$$\binom{n+k-1}{k-1}.$$

□

By the familiar stars-and-bars argument above,

$$D(a,r)=\binom{a+r-1}{r-1}.$$

Then we have already proved the object being counted is not a permutation (since members of A are indistinguishable, only the numbers put into each labelled group matter in this case).

Proof. Now allow n distinguishable types. We then have

$$A_1, A_2, \ldots, A_n$$

with a_i indistinguishable members in A_i, to match the notation in the statement of the problem. For the sake of simplicity, we now impose no constraints on the minimum size and abstention. For every A_i we independently distribute its a_i members amongst r labelled groups. There are exactly

$$D(a_i,r)=\binom{a_i+r-1}{r-1}$$

ways to distribute type A_i. By the multiplication principle,

$$N_0(a_1,\ldots,a_n;r)=\prod_{i=1}^{n}\binom{a_i+r-1}{r-1}.$$

□

Now, we will introduce the minimum size p (note that the abstention requirement is not yet met; it will be addressed later). In other words, every group must contain at least p objects. We unfortunately cannot apply the stars-and-bars argument anymore, because the condition $|G_j|\geq p$ is concerning the total number of objects in G_j, irrespective of their type individually. It is clear that a recursive approach would be better.

Definition 2. We define $F_p(a_1,\ldots,a_n;r)$ as the number of valid distributions of all $a_1+\cdots+a_n$ objects into r labelled groups, with every group containing at least p objects.

We would like to shift our focus onto the very first group. Assume it receives b_1 members from A_1, b_2 members of A_2, or in general,

$$b_n \text{ members of } A_n,$$

necessarily

$$0\leq b_i\leq a_i,$$

and the group-size condition dictates that

$$b_1+\cdots+b_n\geq p.$$

Once these b_i items have been chosen, there is exactly $F_p(a_1 - b_1, \ldots, a_n - b_n; r-1)$ ways to distribute the remaining objects (because the second labelled group in this order would become the 'first group' when we are done with the first group, or just consider with the lens of the above recurrent definition).

These possibilities are guaranteed to be disjoint, and together they exhaust all possibilities (recall the definition of a partition of a set). Therefore

$$F_p(a_1, \ldots, a_n; r) = \sum_{\substack{0 \le b_i < a \\ b_1 + \cdots + b_n \ge p}} F_p(a_1 - b_1, \ldots, a_n - b_n; r-1),$$

where the boundary condition is

$$F_p(a_1, \ldots, a_n; 1) = \begin{cases} 1, & \sum^n a_k \ge p \\ 0, & \sum^n a_k < p \end{cases}.$$

It is also crystal clear that an immediate impossibile condition appears:

$$S < rp \implies F_p = 0.$$

However, once the abstention condition is added, note that this is not an if-and-only-if relation. It can be proved that the above intimidating-looking behemoth of summation is a complete general solution if the last condition is not included.

Now, introduce the abstention condition. For definition, refer to the statement of the problem.

Definition 3. A group is *admissible* when it does not contain members from all n sets.

An alternative consideration is that if an admissible group G receives

$$b_1, \ldots, b_n,$$

then it must satisfy that

$$\exists b_i = 0.$$

Definition 4. Define $F_p^*(a_1, \ldots, a_n; r)$ to be the number of distributions satisfying both

$$\forall G_j, |G_j| \ge p$$

and

$$\forall G, \quad G \text{ is admissible.}$$

As a usual practice, let us examine the very first group again. Its composition must now satisfy the following two conditions:

$$0 \le b_i \le a_i, \quad b_1 + \cdots + b_n \ge p,$$

as well as

$$\min\{b_1, \ldots, b_n\} = 0.$$

It is not hard to infer that

$$F_p^*(a_1, \ldots, a_n; r) = \sum_{\substack{0 \le b_1 \le a_i \\ b_1 + \cdots + b_n \ge p \\ \min_i b_i = 0}} F_p^*(a_1 - b_1, \ldots, a_n - b_n; r-1).$$

For the final group,

$$F_p^*(a_1, \ldots, a_n; 1) = \begin{cases} 1, & \sum^n a_k \ge p, \quad \min_i a_i = 0, \\ 0, & \text{otherwise.} \end{cases}$$

For $n = 1$, notice that F_p and F_p^* are interchangeable.

2.1 Simple Verification

To satisfy the satiety for curiosity of some pedantic readers, we need to make this essay very rigorous and thus it is worth making the logical structure we implicitly employed explicit.

($\Longrightarrow$): We begin with taking any valid arrangement, and observe that it has a unique first-group composition $(b_1, \ldots, b_n)$. The composition satisfies the conditions appearing in the summation.

($\Longleftarrow$): Choose any such composition; any valid arrangement of the remaining objects into the remainign $r-1$ labelled groups produces exactly one valid complete arrangement.

Thus a bijection (recall the definition of bijection) is established to prove that the union is disjoint (with the aid of the addition principle):

$$\{\text{valid complete arrangements}\} \iff \bigcup_{\text{admissible } (b_1, \ldots, b_n)} \{\text{valid arrangements of the remainder}\}.$$

Furthermore, we can **build the inclusion-exclusion** from the same elementary principles without relying on the (in)famous Inclusion-Exclusion Principle. This is a key step to deriving the general formula because the abstention condition happens to have an elegant inclusion-exclusion structure.

Definition 5. Just in this article, a group is said to be *bad* if it fails to satisfy the abstention condition.

In other words, a *bad* group must contain at least one member of every A_i.

Let

$$B_j = \{\text{group } j \text{ is bad}\}.$$

Then we want the arrangement in which $B_1, B_2, \ldots, B_r$ all fail to occur.

Derivation. For an arbitrary arrangement that has exactly m bad groups, inclusion-exclusion assigns it the coefficient

$$\sum_{q=0}^{m} (-1)^q \binom{m}{q}.$$

However,

$$\sum_{q=0}^{m} (-1)^q \binom{m}{q} = (1-1)^m = \begin{cases} 1, & m = 0, \\ 0, & m > 0 \end{cases}.$$

Therefore, the inclusion-exclusion leaves exactly the arrangements with no bad groups. Hence

$$F_p^* = \sum_{q=0}^{r} (-1)^q \binom{r}{q} H_q,$$

where H_q denotes the number of arrangements satisfying the minimum-size condition in which some fixed q labelled groups are all *bad*. The fact that only q matters but not *which* q groups, follows from the symmetry of the labels. □

We now then would like to count H_q from scratch. Suppose groups $G_1, \ldots, G_q$ are required to be bad.

Derivation. For each of these groups, being bad means that it receives at least one member from each A_i. So, for each A_i, we reserve one member for each of the q selected groups, which consumes q members of A_i. Thus we must have

$$\forall i, a_i \geq q$$

or else

$$H_q = 0.$$

After this reservation, the remaining numbers are $a_i - q$. Each selected group already contains n reserved objects.

Therefore, a selected group needs another $p - n$ objects if $p > n$; it needs no addition objects otherwise.

Then, define

$$s = \max(0, p - n);$$

then H_q is precisely the number of ways to distribute the remaining

$$a_1 - q, \ldots, a_n - q$$

objects into r labelled groups such that $G_1, \ldots, G_q$ each receives at least s further objects, while $G_{q+1}, \ldots, G_r$ each receives at least p objects. □

With these on hand, we can define a slightly more general (yet complicated) general function for this.

Let

$$F(a_1, \ldots, a_n; t_1, \ldots, t_r)$$

denote the number of ways to distribute the objects amongst r labelled groups where group j must receive at least t_j objects. Then, by exactly the same first-group decomposition,

$$F(a_1, \ldots, a_n; t_1, \ldots, t_r) = \sum_{\substack{0 \geq b_1 \geq a \\ b_1 + \cdots + b_n \geq t_1}} F(a_1 - b_1, \ldots, a_n - b_n; t_2, \ldots, t_r).$$

When there are no groups left,

$$F(a_1, \ldots, a_n; \emptyset) = \begin{cases} 1, & a_1 = \cdots = a_n = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Now

$$H_q = F\left(a_1 - q, \ldots, a_n - q; \underbrace{s, \ldots, s}_{q}, \underbrace{p, \ldots, p}_{r-q}\right),$$

where $s = \max(0, p - n)$. Therefore the full answer becomes:

$$F_p^*(a_1, \ldots, a_n; r) = \sum_{q=0}^{r} (-1)^q \binom{r}{q} F\left(a_1 - q, \ldots, a_n - q; \underbrace{s, \ldots, s}_{q}, \underbrace{p, \ldots, p}_{r-q}\right) \tag{1}$$

with the q-th term interpreted as 0 whenever $q > \min_i a_i$.

3 Theoretical Summary and Special Cases

Formally, define the desired number $N(n, r, p; \mathbf{a})$ where

$$\mathbf{a} = (a_1, \ldots, a_n),$$

as follows:

$$\begin{cases} n = 1 & N(1, r, p; a_1) = F_p(a_1; r) \\ n > 1 & N(n, r, p; \mathbf{a}) = F_p^*(a_1, \ldots, a_n; r) \\ & \text{where} \quad F_p^*(\mathbf{a}; r) = \sum_{\substack{0 < b_i \le a_i \\ \sum b_i \ge p \\ \exists i : b_i = 0}} F_p^*(\mathbf{a} - \mathbf{b}; r - 1) \\ & \text{with} \quad F_p^*(\mathbf{a}; 1) = 1\left[\sum_i a_i \ge p \wedge \exists i : a_i = 0\right]. \end{cases} \tag{2}$$

Particularly, the case $n = 1$ becomes exceptionally simple, because the abstention rule is practically nonexistent. We are merely distributing a_1 indistinguishable objects into r labelled groups, each with at least p. Define

$$x_j = p + y_j, \qquad y_j \ge 0.$$

It follows that

$$y_1 + \cdots + y_r = a_1 - rp.$$

Repeated usages of the stars-and-bars argument yield

$$F_p(a_1; r) = \begin{cases} \binom{a_1 - rp + r - 1}{r - 1}, & a_1 \ge rp, \\ 0, & a_1 < rp, \end{cases}$$

backed by the general recurrence (the consistency is still present).

3.1 Combinatorial Lenses

We would actually discover that there are three criteria to the problem raised, and each can be given a combinatorial interpretation, ditching all recurrences and algebra aside.

The statement pertaining to depletion implies that every member muts be assigned somewhere, and all a_i members of A_i are exhausted. This is what the equations $0 \le b_i \le a_i$ and the recursive subtraction $a_i - b_i$ enforce.

The statement pertaining to minimum occupancy implies that every group has to satisfy

$$|G_j| \ge p$$

that creates interaction between different types within the same group.

The statement pertaining to abstention implies that every group must satisfy

$$\exists i : \text{group contains no member of } A_i,$$

creating interaction between the groups and types in a different direction.

4 One Example

Example 1. 28 distinguishable candies are distributed amongst 6 kids. Each child must receive at least 3. How many ways to distribute the candies?

To walk through this, we first need to define the quantity we want from the above elegant excursion. Let

$$F(m, r, p)$$

be the number of ways to distribute m distinct candies amongst r labelled children, with every child receiving at least p candies. The problem asks for

$$F(28, 6, 3).$$

Assume the abstention condition does not exist because it is not mentioned in the prompt of the question.

Suppose there is only one child. If there are $m \ge p$ candies, then that the child must receive everything. Observe that there is exactly one distribution

$$F(m, 1, p) = 1;$$

if $m < p$, the requirement cannot be satisfied:

$$F(m, 1, p) = 0.$$

So,

$$F(m, 1, p) = \begin{cases} 1, & m \ge p, \\ 0, & m < p. \end{cases}$$

There is also an immediate impossible condition:

$$m < rp \implies F(m, r, p) = 0.$$

According to the problem,

$$28 \geq 6 \cdot 3 = 18,$$

so it is possible.

Child 1 receives exactly k candies, and because each is distinguishable, there are $\binom{m}{k}$ ways to choose those. For the arrangement to be valid, Child 1 must have $k \geq p$; after choosing Child 1's k candies, there remain $m - k$ candies for the remaining $r - 1$ children. Those remaining children must themselves each receive at least p candies. Hence there are

$$F(m - k, r - 1, p)$$

ways to distribute the remainder. By the multiplication principle, for this particular k there are

$$\binom{m}{k} F(m - k, r - 1, p)$$

arrangements. Finally, k could be anything from p to m, so by the addition principle,

$$F(m, r, p) = \sum_{k=p}^{m} \binom{m}{k} F(m - k, r - 1, p).$$

We just want the actual numerical answer to the above recurrence relation, so by using C++ (disclaimer: not sponsored by Stroustrup), one can get

$$F(28, 6, 3) = 2\,209\,707\,853\,944\,675\,804\,720.$$

As a copmarison, the unrestricted number would have been

$$6^{28} = 6\,140\,942\,214\,464\,815\,497\,216.$$

5 Conclusion

It can be seen amazingly that in combinatorics, that answers are often astronomical despite the simple uses of basic arithmetic. This is indeed the beauty in it – apart from the possibilities measured by figures, the approach to solving combinatorial problems often would be completely different if one constraint is added or changed. For example, the above prose would be completely useless if the abstention policy is changed to 'abstaining from at least any *2* sets'. This actually leaves a vast and good ground for readers to explore, and complex problems can be created (occasionally cannot be solved) by stacking up simple conditions. In a sense, this is similar to the beloved Number Theory.